

HSC GENERATION 1 — REVISION 12

MASTER ENGINEERING DOCUMENT TREE

Documents 001 – 500 | Sequential Numbering

Baseline: 26,000 kg | Main Rotor 26.0 m | Tail Rotor 4.0 m | Capsule 7,000 kg

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This Master Document Tree defines the controlled list of engineering documents required to mature HSC Generation 1 Rev12 from conceptual baseline toward analysis, design, verification and licensing readiness. Each entry has a sequential number (1–500), document name, and document type. Numbers are continuous with no gaps.

Document types include: Master Specification, System Requirements, Architecture Description, Design Description, Interface Control, Analysis Report, Safety Analysis, Test Plan, Test Spec, Mass Budget, Trade Study, IP Description, Licensing Package, Procedure, Checklist, and related controlled types.

Documents 001 – 022

No.	Document Name	Document Type
1	HSC-00 Comprehensive Engineering Baseline Rev12	Master Specification
2	HSC System Requirements Specification (SyRS)	System Requirements
3	HSC System Architecture Description	Architecture Description
4	HSC Configuration Management Plan	Management Plan
5	HSC Master Document List Control Procedure	Procedure
6	HSC Interface Control Master Index	Interface Control
7	HSC Risk Management Plan	Management Plan
8	HSC Safety Management Plan	Safety Plan
9	HSC Verification and Validation Master Plan	V&V; Plan
10	HSC Certification Basis Definition (CS-29 Oriented)	Certification Basis
11	HSC Concept of Operations (CONOPS)	CONOPS
12	HSC Mission Profile Definition	Mission Definition
13	HSC Design Assurance Level Allocation	Safety Allocation
14	HSC System Safety Assessment Plan	Safety Plan
15	HSC Functional Hazard Assessment (FHA)	Safety Analysis
16	HSC Preliminary System Safety Assessment (PSSA)	Safety Analysis
17	HSC System Safety Assessment (SSA)	Safety Analysis
18	HSC Common Cause Analysis Plan	Safety Analysis
19	HSC Particular Risk Analysis Plan	Safety Analysis
20	HSC Zonal Safety Analysis Plan	Safety Analysis
21	HSC Mass Properties Control Plan	Mass Properties
22	HSC Top-Level Mass Budget 26 000 kg	Mass Budget

Documents 023 – 044

No.	Document Name	Document Type
23	HSC Component Mass Breakdown Report	Mass Report
24	HSC Center of Gravity Envelope Definition	Mass Properties
25	HSC Inertia Tensor Estimation Report	Mass Properties
26	HSC Structural Design Criteria	Design Criteria
27	HSC Airframe Primary Structure Design Description	Design Description
28	HSC Fuselage Load Path Analysis	Structural Analysis
29	HSC Tail Boom Structural Analysis	Structural Analysis
30	HSC Engine Mount Structure Design	Design Description
31	HSC Transmission Support Structure Design	Design Description
32	HSC Rotor Mast Support Integration Structure	Design Description
33	HSC External Skins and Fairings Design	Design Description
34	HSC Doors Access Panels and Hinges Design	Design Description
35	HSC Firewalls and Structural Fire Barriers	Design Description
36	HSC Structural Fittings Lugs and Fasteners Spec	Specification
37	HSC Cabin Floor Outside Capsule Boundary	Design Description
38	HSC Airframe Fatigue and Damage Tolerance Plan	Structural Analysis
39	HSC Airframe Static Strength Analysis	Structural Analysis
40	HSC Airframe Dynamic Loads Analysis	Structural Analysis
41	HSC Crashworthiness Criteria Airframe	Design Criteria
42	HSC Materials Selection Report — Airframe	Materials Report
43	HSC Composite Structures Design Allowables	Materials Report
44	HSC Metallic Structures Design Allowables	Materials Report

Documents 045 – 066

No.	Document Name	Document Type
45	HSC Corrosion Protection and Finishing Spec	Specification
46	HSC Manufacturing Tolerances Airframe	Manufacturing Spec
47	HSC Structural Interface Control Document Cap-AF	Interface Control
48	HSC Ground Handling and Tiedown Provisions	Design Description
49	HSC Maintenance Access and Inspection Provisions	Design Description
50	HSC Structural Weight Optimization Trade Study	Trade Study
51	HSC Main Rotor System Requirements	Requirements Spec
52	HSC Main Rotor 26.0 m Design Description	Design Description
53	HSC Main Rotor Blade Aerodynamic Design	Design Description
54	HSC Main Rotor Blade Structural Design	Design Description
55	HSC Main Rotor Hub and Retention Design	Design Description
56	HSC Main Rotor Mast Design	Design Description
57	HSC Main Rotor Swashplate and Pitch Control	Design Description
58	HSC Main Rotor Pitch Links and Control Rods	Design Description
59	HSC Main Rotor Vibration and Dynamic Control	Design Description
60	HSC Main Rotor RPM and Tip Speed Study	Analysis Report
61	HSC Main Rotor Solidity and Twist Optimization	Trade Study
62	HSC Main Rotor Airfoil Family Selection	Trade Study
63	HSC Main Rotor Disk Loading Performance Study	Performance Analysis
64	HSC Main Rotor Hover Power Required Analysis	Performance Analysis
65	HSC Main Rotor Autorotation Performance Analysis	Performance Analysis
66	HSC Main Rotor Loads Analysis	Structural Analysis

Documents 067 – 088

No.	Document Name	Document Type
67	HSC Main Rotor Fatigue Life Assessment	Structural Analysis
68	HSC Main Rotor Ice Protection Concept	Design Description
69	HSC Main Rotor Balancing and Tracking Spec	Specification
70	HSC Tail Rotor System Requirements	Requirements Spec
71	HSC Tail Rotor 4.0 m Design Description	Design Description
72	HSC Tail Rotor Blade Design	Design Description
73	HSC Tail Rotor Hub and Retention Design	Design Description
74	HSC Tail Rotor Gearbox Design	Design Description
75	HSC Tail Rotor Drive Shafting Design	Design Description
76	HSC Tail Rotor Controls and Pitch Mechanism	Design Description
77	HSC Tail Rotor Anti-Torque Authority Analysis	Performance Analysis
78	HSC Tail Rotor Loads and Fatigue Analysis	Structural Analysis
79	HSC Rotor System Interface Control Document	Interface Control
80	HSC Rotor Emergency Release Feasibility Study	Trade Study
81	HSC Propulsion System Requirements	Requirements Spec
82	HSC Turboshift Engine Selection Study	Trade Study
83	HSC Engine Installation Design Description	Design Description
84	HSC Engine Mounts and Vibration Isolation	Design Description
85	HSC Engine Control System Interface Spec	Interface Control
86	HSC Engine Exhaust and Hot Section Installation	Design Description
87	HSC One-Engine-Inoperative Performance Analysis	Performance Analysis
88	HSC Hot and High Performance Analysis	Performance Analysis

Documents 089 – 110

No.	Document Name	Document Type
89	HSC Main Gearbox Combining Transmission Design	Design Description
90	HSC Intermediate and Reduction Gearbox Design	Design Description
91	HSC Drive Shafts and Couplings Design	Design Description
92	HSC Transmission Lubrication and Cooling System	Design Description
93	HSC Transmission Thermal and Torque Limits Analysis	Analysis Report
94	HSC Fuel System Requirements Specification	Requirements Spec
95	HSC Fuel Tank Design Crash Fire Resistant	Design Description
96	HSC Fuel Quantity and Usable Volume Analysis	Analysis Report
97	HSC Fuel Pumps Boost and Transfer Design	Design Description
98	HSC Fuel Lines Valves and Fittings Spec	Specification
99	HSC Fuel Isolation and Emergency Shutoff Design	Design Description
100	HSC Fuel Venting Surge and Inerting Provisions	Design Description
101	HSC Unusable Fuel Allocation Report	Mass Report
102	HSC 1000 km Mission Fuel Model	Performance Analysis
103	HSC Fuel System Fire Hazard Analysis	Safety Analysis
104	HSC Engine Fuel Isolation Coordination with Separation	Interface Control
105	HSC Propulsion System Interface Control Document	Interface Control
106	HSC Engine Separation Feasibility Study	Trade Study
107	HSC Transmission Separation Feasibility Study	Trade Study
108	HSC Propulsion Health Monitoring Requirements	Requirements Spec
109	HSC Propulsion Maintenance Concept	Maintenance Concept
110	HSC Propulsion Weight and Balance Contribution	Mass Report

Documents 111 – 132

No.	Document Name	Document Type
111	HSC Survival Capsule Requirements Specification	Requirements Spec
112	HSC Capsule Geometry and Layout Definition	Design Description
113	HSC Capsule 7000 kg Mass Budget Bottom-Up	Mass Budget
114	HSC Capsule Primary Crashworthy Cell Design	Design Description
115	HSC Capsule Internal Roll-Cage CFRP Design	Design Description
116	HSC Capsule Roll-Cage Joints and Titanium Nodes	Design Description
117	HSC Capsule Outer Shell Fairing Thermal Protection	Design Description
118	HSC Capsule Structural Load Path Analysis	Structural Analysis
119	HSC Capsule Crash Load Analysis	Structural Analysis
120	HSC Capsule Fatigue and Damage Tolerance Analysis	Structural Analysis
121	HSC Capsule Occupant Injury Criteria Definition	Design Criteria
122	HSC Capsule Energy-Absorbing Seats Design	Design Description
123	HSC Capsule Seat Restraint System Spec	Specification
124	HSC Capsule Floor and Secondary Structure Design	Design Description
125	HSC Capsule Emergency Exits Doors Frames Seals	Design Description
126	HSC Capsule Emergency Egress Analysis	Safety Analysis
127	HSC Capsule Independent Emergency Power Design	Design Description
128	HSC Capsule Emergency Battery System Spec	Specification
129	HSC Capsule Life Support Oxygen Environmental	Design Description
130	HSC Capsule Fire Smoke Detection Suppression	Design Description
131	HSC Capsule Water Landing Flotation Sealing	Design Description
132	HSC Capsule Avionics Sensing Communications Locator	Design Description

Documents 133 – 154

No.	Document Name	Document Type
133	HSC Capsule Lighting and Post-Landing Visibility	Design Description
134	HSC Capsule Separation Interface Hardware Design	Design Description
135	HSC Capsule Attachment Fittings Design	Design Description
136	HSC Capsule-to-Airframe Interface Control CAP-AF	Interface Control
137	HSC Capsule Clearance Verification Requirements	Requirements Spec
138	HSC Capsule Post-Landing Stability Analysis	Analysis Report
139	HSC Capsule Asymmetric Failed-Leg Landing Cases	Structural Analysis
140	HSC Capsule Materials Selection CFRP Metallic	Materials Report
141	HSC Capsule Manufacturing and Assembly Concept	Manufacturing Spec
142	HSC Capsule Inspection and Replaceable Hardware	Maintenance Concept
143	HSC Capsule Growth Allowance Management	Mass Report
144	HSC Capsule Environmental Qualification Plan	Test Plan
145	HSC Capsule Crash Test Plan	Test Plan
146	HSC Capsule Occupant Protection Test Plan	Test Plan
147	HSC Eight Landing Supports Requirements	Requirements Spec
148	HSC Eight Landing Supports Design Description	Design Description
149	HSC Landing Support Energy Absorber Design	Design Description
150	HSC Landing Support Load Path Analysis	Structural Analysis
151	HSC Landing Support Asymmetric Failure Analysis	Structural Analysis
152	HSC Landing Support Terrain Interaction Study	Analysis Report
153	HSC Landing Support Ground Clearance Analysis	Analysis Report
154	HSC Landing Support Rotor Downwash Interaction	Analysis Report

Documents 155 – 176

No.	Document Name	Document Type
155	HSC Landing Support Replaceable Crush Elements Spec	Specification
156	HSC Landing Support Interface Control CAP-LG	Interface Control
157	HSC Capsule Parachute System Requirements	Requirements Spec
158	HSC Distributed Multi-Canopy 8-Canopy Architecture	Design Description
159	HSC Capsule Parachute Sizing Analysis	Analysis Report
160	HSC Capsule Parachute Deployment Sequence Design	Design Description
161	HSC Capsule Parachute Reefing and Riser Design	Design Description
162	HSC Capsule Parachute Failure Tolerance Analysis	Safety Analysis
163	HSC Capsule Parachute Inflation Dynamics Study	Analysis Report
164	HSC Capsule Parachute Interface Control CAP-PAR	Interface Control
165	HSC Optional Residual Airframe Parachute Trade Study	Trade Study
166	HSC Multi-Mode Recovery Architecture Description	Architecture Description
167	HSC Controlled Flight Recovery Mode Definition	Design Description
168	HSC Autorotation Recovery Mode Definition	Design Description
169	HSC Selective Mass Hazard Separation Mode	Design Description
170	HSC Capsule Separation Recovery Mode Definition	Design Description
171	HSC Post-Landing Survival Locator Communications	Design Description
172	HSC Rapid Separation System Requirements	Requirements Spec
173	HSC Selective Separation Philosophy Description	Architecture Description
174	HSC State-Dependent Release Logic Specification	Requirements Spec
175	HSC Candidate Separable Components List	Design Description
176	HSC Primary Capsule Release Mechanisms Design	Design Description

Documents 177 – 198

No.	Document Name	Document Type
177	HSC Backup Independent Release Mechanisms Design	Design Description
178	HSC Release Mechanism Locks and Inhibits Design	Design Description
179	HSC Release State Sensing System Design	Design Description
180	HSC Independent Release Power and Signalling	Design Description
181	HSC Release Arm Fire Sequence Specification	Requirements Spec
182	HSC Fuel Energy Isolation Before Release Spec	Requirements Spec
183	HSC Asymmetric Partial Release Detection Design	Design Description
184	HSC Post-Release Trajectory Monitoring Concept	Design Description
185	HSC Capsule Clear Verification Before Parachute	Requirements Spec
186	HSC Failed Release Rollback Fallback Definition	Requirements Spec
187	HSC Six-Second Survival Sequence Development Spec	Requirements Spec
188	HSC Separation Kinematics Analysis	Analysis Report
189	HSC Separation Trajectory Clearance Analysis	Analysis Report
190	HSC Released Component Collision Envelope Analysis	Safety Analysis
191	HSC Separation System Mass Allocation 1000 kg	Mass Report
192	HSC Separation System Interface Control JET-CMD	Interface Control
193	HSC Separation System Interface Control JET-PWR	Interface Control
194	HSC Separation System Interface Control JET-SNS	Interface Control
195	HSC Separation Common Cause Independence Analysis	Safety Analysis
196	HSC Unintended Separation Hazard Analysis	Safety Analysis
197	HSC Separation Mechanism Reliability Target Spec	Requirements Spec
198	HSC Separation Ground Test Plan	Test Plan

Documents 199 – 220

No.	Document Name	Document Type
199	HSC Separation Integrated Demonstration Plan	Test Plan
200	HSC Separation Dynamic Simulation Requirements	Requirements Spec
201	HSC Six-Layer Survival Architecture Description	Architecture Description
202	HSC Layer L1 Automatic Detection Initiation Spec	Requirements Spec
203	HSC Layer L2 Independent Backup Initiation Spec	Requirements Spec
204	HSC Layer L3 Manual Crew Authority Spec	Requirements Spec
205	HSC Layer L4 Selective Physical Separation Spec	Requirements Spec
206	HSC Layer L5 Direct Impact Survival Spec	Requirements Spec
207	HSC Layer L6 Recovery Post-Event Survival Spec	Requirements Spec
208	HSC Safety Layer Independence Assessment Plan	Safety Analysis
209	HSC Deterministic Safety Kernel Requirements	Requirements Spec
210	HSC Safety Kernel Authorization Preconditions Spec	Requirements Spec
211	HSC Safety Kernel Altitude Airspeed Envelope Logic	Design Description
212	HSC Safety Kernel Attitude Rate Limits Logic	Design Description
213	HSC Safety Kernel Rotor Engine Fire State Logic	Design Description
214	HSC Safety Kernel Structural Health Logic	Design Description
215	HSC Safety Kernel Sensor Validity Disagreement Logic	Design Description
216	HSC Safety Kernel Human Authority Override Spec	Requirements Spec
217	HSC Safety Kernel Software Design Assurance	Software Spec
218	HSC Emergency State Machine Specification	Requirements Spec
219	HSC Emergency State Transition Matrix	Design Description
220	HSC State-Dependent Separation Set Definition	Design Description

Documents 221 – 242

No.	Document Name	Document Type
221	HSC Intelligent Monitoring Prediction Boundary Spec	Requirements Spec
222	HSC AI Decision Support Constraints Specification	Requirements Spec
223	HSC Health Monitoring System Requirements	Requirements Spec
224	HSC Digital Twin Estimation Concept	Design Description
225	HSC Predictive Fault Detection Concept	Design Description
226	HSC Immutable Emergency Event Record Spec	Requirements Spec
227	HSC Avionics System Requirements Specification	Requirements Spec
228	HSC Flight Control Computers Design Description	Design Description
229	HSC Flight Control Actuators Servo Spec	Specification
230	HSC Navigation Air Data Flight Sensors Spec	Specification
231	HSC Communications Mission Avionics Spec	Specification
232	HSC Electrical Generation Distribution Design	Design Description
233	HSC Electrical Wiring Harness Connectors Spec	Specification
234	HSC Battery Emergency Electrical Interfaces	Design Description
235	HSC Hydraulic System Design Description	Design Description
236	HSC Thermal Management Cooling Design	Design Description
237	HSC Environmental Control Ventilation Design	Design Description
238	HSC Fire Detection Warning Suppression Interfaces	Design Description
239	HSC Health Monitoring Data Recording Design	Design Description
240	HSC Lighting Service Systems Design	Design Description
241	HSC Avionics Interface Control AV-CTRL Safety Kernel	Interface Control
242	HSC Electromagnetic Compatibility Plan	EMC Plan

Documents 243 – 264

No.	Document Name	Document Type
243	HSC Lightning Protection Design	Design Description
244	HSC Software Configuration Integrity Controls	Software Spec
245	HSC Avionics Weight Allocation Report	Mass Report
246	HSC Avionics Failure Modes Effects Analysis	Safety Analysis
247	HSC Avionics Environmental Qualification Plan	Test Plan
248	HSC Human Machine Interface Emergency Controls	Design Description
249	HSC Crew Procedures Emergency Separation Manual	Procedure
250	HSC Training Requirements Emergency Systems	Training Spec
251	HSC Performance Requirements Specification	Requirements Spec
252	HSC Power Required Power Available Analysis	Performance Analysis
253	HSC Hover Ceiling Hot High Analysis	Performance Analysis
254	HSC Rate of Climb Cruise Performance Analysis	Performance Analysis
255	HSC Cruise Drag Polar Estimation	Performance Analysis
256	HSC Mission Fuel Flow Analysis 1000 km	Performance Analysis
257	HSC Reserve Policy Definition	Requirements Spec
258	HSC Rotor Vibration Dynamic Stability Analysis	Analysis Report
259	HSC 6-DOF Emergency Dynamics Model Spec	Requirements Spec
260	HSC 6-DOF Aircraft Pre-Release Dynamics	Analysis Report
261	HSC 6-DOF Capsule Post-Separation Dynamics	Analysis Report
262	HSC 6-DOF Released Component Trajectory Models	Analysis Report
263	HSC 6-DOF Partial Asymmetric Release Cases	Analysis Report
264	HSC 6-DOF Parachute Inflation Transient Loads	Analysis Report

Documents 265 – 286

No.	Document Name	Document Type
265	HSC 6-DOF Ground Contact Kinematics Slope Terrain	Analysis Report
266	HSC 6-DOF Occupant Acceleration Pulse Screening	Analysis Report
267	HSC 6-DOF Model Uncertainty Sensitivity Report	Analysis Report
268	HSC Flight Envelope Definition Normal Emergency	Requirements Spec
269	HSC Handling Qualities Requirements	Requirements Spec
270	HSC Stability and Control Analysis	Analysis Report
271	HSC Aerodynamic Database Development Plan	Analysis Plan
272	HSC Wind Tunnel Test Plan if Applicable	Test Plan
273	HSC CFD Analysis Plan Rotor Airframe	Analysis Plan
274	HSC Noise Certification Assessment Plan	Certification Plan
275	HSC Performance Flight Test Plan Outline	Test Plan
276	HSC Operating Fluids Consumables Mass Report	Mass Report
277	HSC Growth Configuration Reserve Management	Mass Report
278	HSC Weight Empty Definition Report	Mass Report
279	HSC Payload Occupancy Accommodation Study	Trade Study
280	HSC Cabin Ergonomics 10-12 Occupants Study	Design Description
281	HSC Geometry Overall Length Width Height Closure	Design Description
282	HSC Ground Clearance Rotor Downwash Study	Analysis Report
283	HSC External Geometry Aerodynamic Fairing Design	Design Description
284	HSC Fixed Landing Support Permanent Drag Impact	Analysis Report
285	HSC Performance Comparison vs Conventional Rotorcraft	Trade Study
286	HSC Hover Endurance Derivation After Power Closure	Performance Analysis

Documents 287 – 308

No.	Document Name	Document Type
287	HSC OEI Safe Operating Envelope Definition	Requirements Spec
288	HSC Autorotation Entry and Flare Analysis	Performance Analysis
289	HSC Transmission Thermal Limit Performance Impact	Analysis Report
290	HSC Density Altitude Performance Sensitivity	Performance Analysis
291	HSC Flight Manual Performance Section Outline	Documentation Spec
292	HSC Performance Model Validation Plan	V&V; Plan
293	HSC Simulation Facility Requirements 6-DOF	Requirements Spec
294	HSC Pilot-in-the-Loop Simulation Plan Outline	Test Plan
295	HSC Handling Qualities Simulation Matrix	Test Plan
296	HSC Emergency Procedure Simulation Matrix	Test Plan
297	HSC Performance Uncertainty Quantification	Analysis Report
298	HSC Mission Effectiveness Analysis Outline	Analysis Report
299	HSC Operational Suitability Assessment Outline	Analysis Report
300	HSC Performance Closure Gate G1 Checklist	Checklist
301	HSC Verification Matrix Master	V&V; Plan
302	HSC G0 Architecture Freeze Checklist	Checklist
303	HSC G1 Analytical Closure Checklist	Checklist
304	HSC G2 Subsystem Demonstration Plan	Test Plan
305	HSC G3 Integrated Ground Demonstration Plan	Test Plan
306	HSC G4 Dynamic 6-DOF Simulation Plan	Test Plan
307	HSC G5 Structural Crash Testing Plan	Test Plan
308	HSC G6 Recovery System Testing Plan	Test Plan

Documents 309 – 330

No.	Document Name	Document Type
309	HSC G7 Flight Development Plan Outline	Test Plan
310	HSC G8 Certification Qualification Package Outline	Certification Plan
311	HSC Release Mechanism Subsystem Test Spec	Test Spec
312	HSC Lock Inhibit Subsystem Test Spec	Test Spec
313	HSC Energy Absorber Subsystem Test Spec	Test Spec
314	HSC Parachute Component Test Spec	Test Spec
315	HSC Sensor Independent Power Test Spec	Test Spec
316	HSC Capsule Structural Static Test Spec	Test Spec
317	HSC Capsule Crash Test Spec	Test Spec
318	HSC Seat Occupant Protection Test Spec	Test Spec
319	HSC Landing Support Drop Test Spec	Test Spec
320	HSC Capsule Attachment Fitting Test Spec	Test Spec
321	HSC Separation Timing Reliability Test Spec	Test Spec
322	HSC Safety Kernel Software Test Spec	Test Spec
323	HSC End-to-End Survival Sequence Ground Test	Test Spec
324	HSC Partial Asymmetric Separation Test Cases	Test Spec
325	HSC Parachute Deployment Inflation Test Spec	Test Spec
326	HSC Canopy Failure Case Test Spec	Test Spec
327	HSC Post-Landing Stability Rescue Access Test	Test Spec
328	HSC Environmental Qualification Test Matrix	Test Plan
329	HSC EMI Lightning Test Plan Outline	Test Plan
330	HSC Fire Fuel Isolation Test Plan Outline	Test Plan

Documents 331 – 352

No.	Document Name	Document Type
331	HSC Test Instrumentation Data Acquisition Spec	Specification
332	HSC Test Safety Plan Ground Demonstrations	Safety Plan
333	HSC Flight Test Safety Plan Outline	Safety Plan
334	HSC Evidence Package Structure Definition	Documentation Spec
335	HSC Requirements Traceability Matrix	Traceability Matrix
336	HSC Verification Evidence Register	Register
337	HSC Nonconformance and Deviation Procedure	Procedure
338	HSC Test Report Template Standard	Template
339	HSC Analysis Report Template Standard	Template
340	HSC Design Review Entry Exit Criteria	Procedure
341	HSC Preliminary Design Review Plan	Review Plan
342	HSC Critical Design Review Plan	Review Plan
343	HSC Test Readiness Review Plan	Review Plan
344	HSC Flight Readiness Review Outline	Review Plan
345	HSC Configuration Audit Plan	Audit Plan
346	HSC Software Design Assurance Plan	Software Plan
347	HSC Hardware Design Assurance Plan	Hardware Plan
348	HSC Supplier Quality Requirements Spec	Specification
349	HSC Acceptance Test Procedure Outline	Procedure
350	HSC Continuing Airworthiness Concept Outline	Maintenance Concept
351	HSC IP Element 0001 Occupied Survival Boundary	IP Description
352	HSC IP Element 0002 State-Dependent Selective Separation	IP Description

Documents 353 – 374

No.	Document Name	Document Type
353	HSC IP Element 0003 Deterministic Safety Kernel	IP Description
354	HSC IP Element 0004 Six-Layer Survival Architecture	IP Description
355	HSC IP Element 0005 Eight-Support Impact Attenuation	IP Description
356	HSC IP Element 0006 Multi-Mode Recovery Chain	IP Description
357	HSC IP Element 0007 Partial Asymmetric Release Management	IP Description
358	HSC IP Element 0008 Residual Airframe Recovery Option	IP Description
359	HSC IP Element 0009 Immutable Emergency Event Record	IP Description
360	HSC Prior Art Search Plan Patent Landscape	IP Plan
361	HSC Freedom-to-Operate Assessment Outline	IP Plan
362	HSC Claim Drafting Guidance Package Outline	IP Plan
363	HSC Licensing Data Room Contents List	Licensing Package
364	HSC Licensee Due Diligence Questionnaire	Licensing Package
365	HSC Strategic Licensing Path Description	Licensing Package
366	HSC Technology Platform Family Concept Utility Rescue	Architecture Description
367	HSC NDA Template Technical Evaluation Scope	Legal Template
368	HSC Controlled Data Room Access Procedure	Procedure
369	HSC Term Sheet Outline Territory Field of Use	Licensing Package
370	HSC Engineering Development Agreement Outline	Licensing Package
371	HSC Program Work Breakdown Structure Level 1-3	Program Plan
372	HSC Integrated Master Schedule Outline	Program Plan
373	HSC Cost Estimating Framework Outline	Program Plan
374	HSC Resource and Skills Requirements Outline	Program Plan

Documents 375 – 396

No.	Document Name	Document Type
375	HSC Make-Buy Strategy Outline	Program Plan
376	HSC Supplier Engagement Plan Outline	Program Plan
377	HSC Risk Register Initial Top Risks	Risk Register
378	HSC Opportunity Register Architecture IP	Register
379	HSC Stakeholder Communication Plan Outline	Management Plan
380	HSC Document Control and Numbering Procedure	Procedure
381	HSC Change Control Board Procedure	Procedure
382	HSC Lessons Learned Capture Procedure	Procedure
383	HSC Export Control Classification Outline	Compliance Spec
384	HSC Security Classification Handling Outline	Compliance Spec
385	HSC Environmental Regulatory Assessment Outline	Compliance Spec
386	HSC Occupational Health Safety Development Site	Safety Plan
387	HSC Prototype Build Philosophy Outline	Manufacturing Spec
388	HSC Tooling and Ground Support Equipment Outline	Manufacturing Spec
389	HSC Logistics Support Analysis Outline	Maintenance Concept
390	HSC Spares Provisioning Concept Outline	Maintenance Concept
391	HSC Technical Publications Plan Outline	Documentation Spec
392	HSC Training System Development Outline	Training Spec
393	HSC Operator Transition Plan Outline	Program Plan
394	HSC Metrics and KPI Dashboard Definition	Management Plan
395	HSC Independent Review Board Charter Outline	Management Plan
396	HSC External Advisory Technical Panel Outline	Management Plan

Documents 397 – 418

No.	Document Name	Document Type
397	HSC Academic Collaboration Framework Outline	Program Plan
398	HSC Public Disclosure and Marketing Claims Policy	Policy
399	HSC Document 00 to Document Tree Traceability	Traceability Matrix
400	HSC Master Document Tree Control Sheet 001-500	Control Sheet
401	HSC CAP-AF Structural Attachment Detailed Design	Design Description
402	HSC CAP-PAR Parachute Attachment Detailed Design	Design Description
403	HSC CAP-LG Landing Support Attachment Detailed Design	Design Description
404	HSC FUEL-ISO Isolation Valve Sequence Spec	Requirements Spec
405	HSC REC-LOC Locator Antenna Strategy Design	Design Description
406	HSC JET-CMD Command Path Independence Design	Design Description
407	HSC JET-PWR Segregated Power Architecture Design	Design Description
408	HSC JET-SNS Release State Sensor Redundancy Design	Design Description
409	HSC Capsule Internal Environment Post-Separation Spec	Requirements Spec
410	HSC Capsule Emergency Oxygen Duration Analysis	Analysis Report
411	HSC Capsule Emergency Communications Range Analysis	Analysis Report
412	HSC Capsule Locator Beacon Specification	Specification
413	HSC Capsule Post-Landing Lighting Specification	Specification
414	HSC Capsule Fire Suppression Agent Selection Study	Trade Study
415	HSC Capsule Smoke Detection Zoning Design	Design Description
416	HSC Capsule Thermal Protection Materials Study	Trade Study
417	HSC Capsule Waterproofing Sealing Design	Design Description
418	HSC Capsule Emergency Exit Marking and Lighting	Design Description

Documents 419 – 440

No.	Document Name	Document Type
419	HSC Capsule Occupant Restraint Dynamic Analysis	Structural Analysis
420	HSC Capsule Seat Energy Absorber Stroke Analysis	Structural Analysis
421	HSC Eight-Support Load Distribution Optimization	Analysis Report
422	HSC Landing Support Soft Soil Penetration Study	Analysis Report
423	HSC Landing Support Slope Landing Stability Study	Analysis Report
424	HSC Landing Support Combined Load Cases Matrix	Structural Analysis
425	HSC Main Rotor Blade Mass Estimate 980 kg Closure	Mass Report
426	HSC Tail Rotor Blade Mass Estimate 145 kg Closure	Mass Report
427	HSC Engine Installed Mass Estimate 800 kg Each	Mass Report
428	HSC Main Gearbox Mass Estimate 750 kg Closure	Mass Report
429	HSC Usable Fuel Mass 2450 kg Mission Closure	Mass Report
430	HSC Separation Hardware Mass 1000 kg Closure	Mass Report
431	HSC Avionics Group Mass 1600 kg Closure	Mass Report
432	HSC Airframe Group Mass 5000 kg Closure	Mass Report
433	HSC Growth Reserve 2000 kg Allocation Rules	Mass Report
434	HSC CG Shift During Staged Separation Analysis	Mass Properties
435	HSC Inertia Change During Staged Separation Analysis	Mass Properties
436	HSC Post-Separation Capsule CG and Inertia Report	Mass Properties
437	HSC Rotor Gyroscopic Effects on Separation Study	Analysis Report
438	HSC Tail Rotor Separation Dynamics Study	Analysis Report
439	HSC Engine Separation Dynamics Study	Analysis Report
440	HSC Fuel Tank Separation Dynamics Study	Analysis Report

Documents 441 – 462

No.	Document Name	Document Type
441	HSC Main Rotor Non-Universal Jettison Rationale	Analysis Report
442	HSC Clearance Envelope Capsule vs Released Hardware	Analysis Report
443	HSC Parachute vs Released Hardware Collision Study	Safety Analysis
444	HSC Minimum Recovery Altitude Definition Analysis	Analysis Report
445	HSC Maximum Dynamic Pressure Separation Limit	Requirements Spec
446	HSC Attitude Rate Limits for Safe Separation	Requirements Spec
447	HSC Rotor State Criteria for Separation Authorization	Requirements Spec
448	HSC Fire State Criteria for Separation Authorization	Requirements Spec
449	HSC Structural Failure State Detection Criteria	Requirements Spec
450	HSC Sensor Disagreement Handling Logic Spec	Requirements Spec
451	HSC Power Availability Criteria for Release Recovery	Requirements Spec
452	HSC Communication Loss Safe State Definition	Requirements Spec
453	HSC Manual Inhibit Path Independence Verification	Safety Analysis
454	HSC Automatic Path Independence Verification	Safety Analysis
455	HSC Common Wiring Route Avoidance Design Rules	Design Criteria
456	HSC Common Power Bus Avoidance Design Rules	Design Criteria
457	HSC Common Structure Vulnerability Assessment	Safety Analysis
458	HSC Common Software Mode Avoidance Rules	Software Spec
459	HSC Fire Zone Segregation Design Rules	Design Criteria
460	HSC EMI Lightning Common Cause Mitigation	Design Description
461	HSC Manufacturing Process Spec Capsule CFRP	Manufacturing Spec
462	HSC Manufacturing Process Spec Airframe	Manufacturing Spec

Documents 463 – 484

No.	Document Name	Document Type
463	HSC Assembly Sequence Capsule into Airframe	Manufacturing Spec
464	HSC Separation Mechanism Installation Procedure	Procedure
465	HSC Functional Check Procedure Separation System	Procedure
466	HSC Functional Check Procedure Safety Kernel	Procedure
467	HSC Functional Check Procedure Capsule Systems	Procedure
468	HSC Ground Run Procedure Propulsion Rotor	Procedure
469	HSC Pre-Flight Inspection Emergency Systems	Procedure
470	HSC Post-Event Inspection and Data Recovery	Procedure
471	HSC Maintenance Manual Chapter Emergency Systems Outline	Documentation Spec
472	HSC Illustrated Parts Catalog Emergency Hardware Outline	Documentation Spec
473	HSC Troubleshooting Guide Separation System Outline	Documentation Spec
474	HSC Service Bulletin Process Outline	Procedure
475	HSC Airworthiness Directive Response Process Outline	Procedure
476	HSC Reliability Prediction Separation Mechanisms	Reliability Analysis
477	HSC Reliability Prediction Capsule Systems	Reliability Analysis
478	HSC Maintainability Analysis Emergency Systems	Maintainability Analysis
479	HSC Human Factors Assessment Emergency Controls	Human Factors
480	HSC Crew Workload Assessment Emergency Sequence	Human Factors
481	HSC Passenger Briefing Emergency Capsule Concept	Training Spec
482	HSC Rescue Services Interface Guidance Outline	Operational Spec
483	HSC Accident Investigation Data Preservation Spec	Requirements Spec
484	HSC Simulation Validation Correlation Plan	V&V; Plan

Documents 485 – 500

No.	Document Name	Document Type
485	HSC Model-Based Systems Engineering Approach Outline	Architecture Description
486	HSC Digital Thread Configuration Data Strategy	Management Plan
487	HSC Cybersecurity Considerations Avionics Outline	Security Spec
488	HSC Supply Chain Risk Assessment Outline	Risk Register
489	HSC Obsolescence Management Plan Outline	Management Plan
490	HSC End-of-Life and Disposal Concept Outline	Environmental Spec
491	HSC Sustainability Materials and Processes Outline	Environmental Spec
492	HSC Variant Derivative Control Procedure Outline	Procedure
493	HSC Single Rotor Baseline Freeze Criteria Rev12	Checklist
494	HSC Open Items Closure Tracking Register	Register
495	HSC Critical Path to First Ground Demonstration	Program Plan
496	HSC Critical Path to First Flight Development	Program Plan
497	HSC Licensing Readiness Checklist	Checklist
498	HSC Document Tree Completeness Audit Sheet	Audit Sheet
499	HSC Cross-Reference Matrix Doc00 to Tree 001-500	Traceability Matrix
500	HSC Master Document Tree Sign-Off Record Rev12	Control Sheet

END OF MASTER DOCUMENT TREE — HSC-00 REV12 | DOCUMENTS 001–500

Sequential numbering verified: 1 through 500 continuous | Founder: Amirbahador Atai